

Weak Interactions

First exp evidence $n \rightarrow p e^- + x$

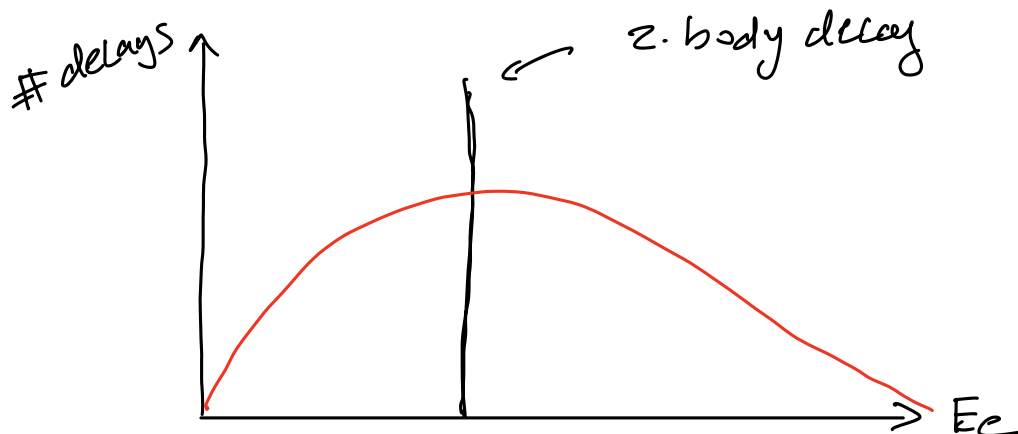
$$n \rightarrow p e^- \quad Q > 0$$

$$p \leftarrow \overset{n}{\bigcirc} \rightarrow e^- \quad n \text{ rest frame.}$$

$$\vec{p}_{e^-} = -\vec{p}_p$$

$$m_n = E_p + E_e$$

$\Rightarrow$ monochromatic decay



β -decay

$$n \rightarrow p + e^- + \bar{\nu}_e$$

Indirect evidence for neutrino

Reines-Cowan experiment

$$n \rightarrow p + e^- + x \quad \Rightarrow \quad x + p \rightarrow n + e^+$$

$$\bigcirc \equiv x \Rightarrow \text{interact with a target}$$

$$\Rightarrow x = \bar{\nu}_e$$

only $n + e^+$ events.
no $n + \gamma^+$ events.

$$\overset{A}{Z} X \rightarrow \overset{A}{Z-1} Y + e^+ + \nu_e \quad \beta^+ \text{ decay.}$$

$$\overset{A}{Z} X \rightarrow \overset{A}{Z+1} Y + e^- + \bar{\nu}_e \quad \beta^- \text{ decay.}$$

1) how to prove $\nu_e \neq \nu_\mu \neq \nu_\tau$

2) $\nu_e \neq \bar{\nu}_e$?

$$\textcircled{C} \quad n \rightarrow \bar{n} \quad n \neq \bar{n}$$

$$\textcircled{C} \quad \pi^0 \rightarrow \pi^0$$

$$\textcircled{C} \quad \gamma \rightarrow \gamma$$

$$\textcircled{C} \quad \nu \rightarrow \bar{\nu} \quad ?$$

$$\nu \rightarrow \nu \quad ?$$

$\nu \neq \bar{\nu}$ Dirac neutrino

$\nu \equiv \bar{\nu}$ Majorana neutrino

open question under investigation.

$$\begin{pmatrix} \ell^- \\ \nu_\ell \end{pmatrix} \quad L_\ell = -1$$

$$\begin{pmatrix} \bar{\nu}_\ell \\ \ell^+ \end{pmatrix} \quad L_\ell = +1$$

$$\ell = e, \mu, \tau$$

3 different lepton flavors : conserved quantities in QED, QCD, weak

Exp evidence $\nu_e \neq \nu_\mu$

AGS @ Brookhaven

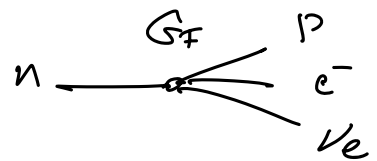
$$n \rightarrow p + e^- + X$$

$$Q > 0$$

$$n \rightarrow p + \mu^- + X$$

$$Q < 0$$

$p + Be \rightarrow$ hadrons.



$$p + Be \rightarrow \text{pions} + X$$

$$\pi^+ \rightarrow \mu^+ \nu_\mu$$

$$\pi^- \rightarrow \mu^- \bar{\nu}_\mu$$

β decay of pions.

$$p + Be \rightarrow \pi^\pm + X$$

$$L \rightarrow \mu^\pm \nu_\mu / \bar{\nu}_\mu$$

$$\Rightarrow \nu_\mu / \bar{\nu}_\mu + X \rightarrow \mu^\pm + \gamma ?$$

beam of neutrinos against target

$$\text{Count } \# \quad e^{+/-} + X$$

$$\# \quad \mu^{+/-} + X$$

events.

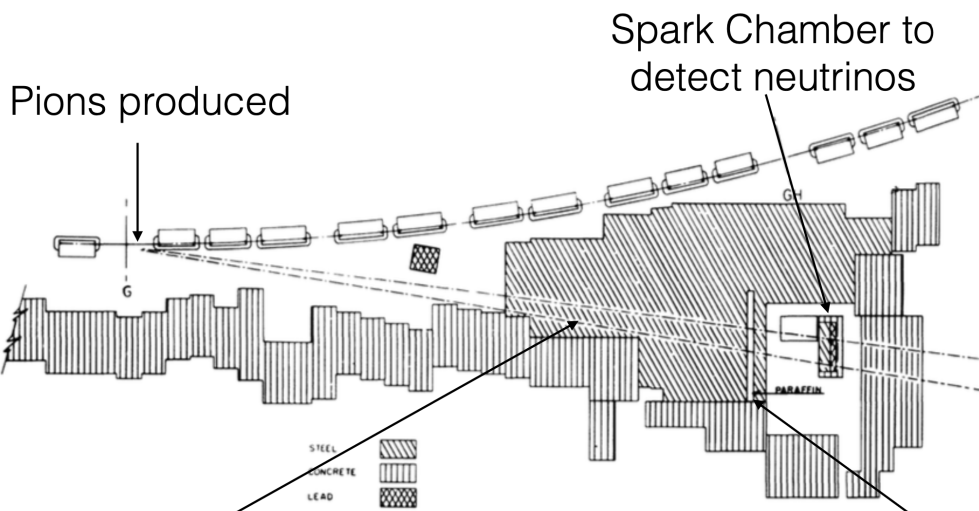


FIG. 1. Plan view of AGS neutrino experiment.

15 MeV protons
on Be target
 3.5×10^{17}
protons on target

$$p+p \rightarrow p+p + \pi^+ + \pi^-$$

? 10 module(s) of $2T$ each

Steel shield stops strongly interacting particles

$$p+p \rightarrow \pi^+ + \pi^- + X$$

$$\hookrightarrow \mu^\pm + \nu_\mu / \bar{\nu}_\mu$$

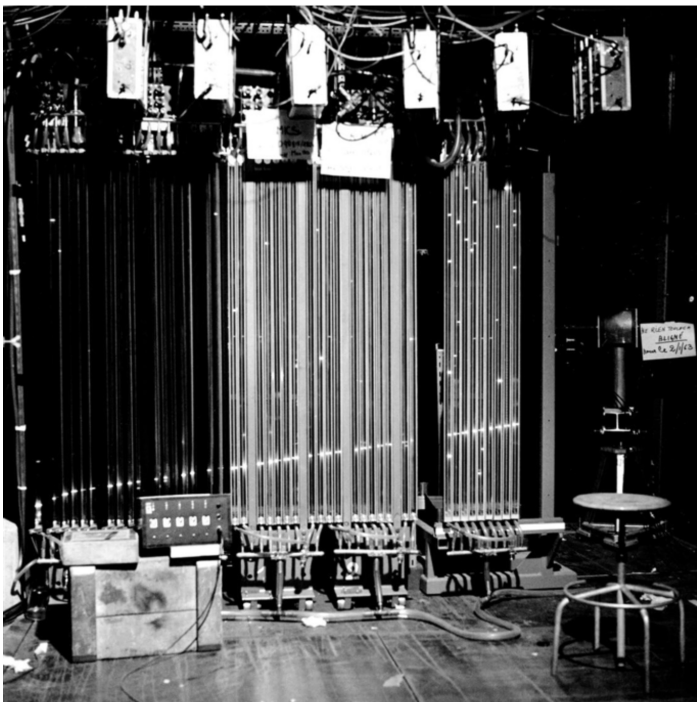
$$\hookrightarrow \nu_\mu / \bar{\nu}_\mu + X \rightarrow \mu^\pm + X$$

$$\tau_\mu = 2.2 \times 10^{-6} \text{ s.}$$

$$e^\pm + \nu_e / \bar{\nu}_e$$

$$\hookrightarrow \nu_e / \bar{\nu}_e + X \rightarrow e^\pm + X$$

Background
suppressed by 10^{-4}



$\mu^\pm$: tracks with
ionization

$e^\pm$: EM showers

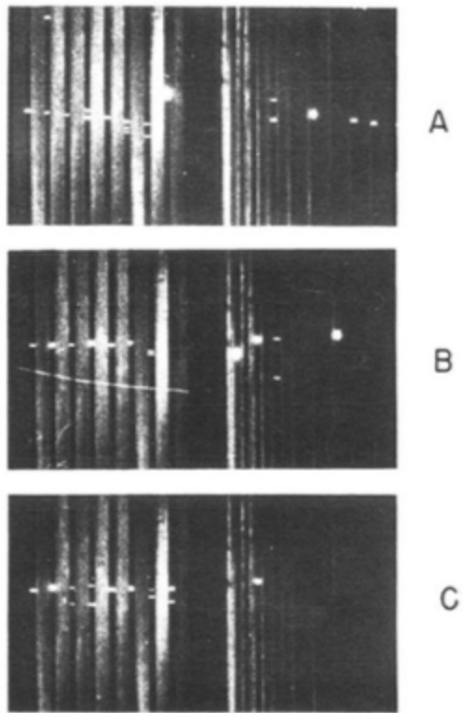


FIG. 8. 400-MeV electrons from the Cosmotron.

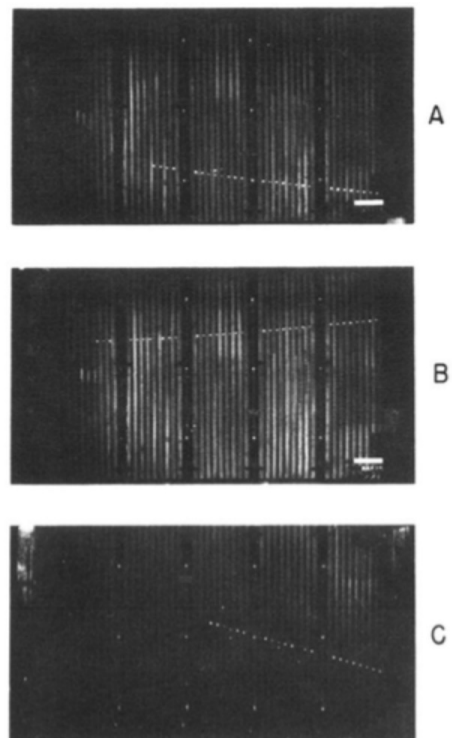


FIG. 5. Single muon events. (A) $p_\mu > 540$ MeV and δ ray indicating direction of motion (neutrino beam incident from left); (B) $p_\mu > 700$ MeV/c; (C) $p_\mu > 440$ with δ ray.

$\mu^- + X \rightarrow 34$ events

6 EM showers in agreement with expected background
no production of $e^\pm$ in $X_\mu + X$ interaction.

$$\pi^+ \rightarrow \mu^+ \nu_\mu$$

$$\pi^- \rightarrow \mu^- \bar{\nu}_\mu$$

$$\pi^+ \rightarrow e^+ \nu_e$$

$$\pi^- \rightarrow e^- \bar{\nu}_e$$

$$Q = m_\pi - m_\mu \approx 26 \text{ MeV}$$

$$\approx 10^{-4} \text{ compared to muons.}$$

$$Q = m_\pi - m_e \approx 140 \text{ MeV}$$

Exp. evidence $\nu_e \neq \nu_\mu$

ν_τ : DONUT @ FERMILAB

$$\nu_\tau \neq \nu_e \neq \nu_\mu.$$

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}$$

Phenomenology of weak decays

$n \rightarrow p e^- \bar{\nu}_e$ semileptonic decay of hadrons

$\pi^+ \rightarrow \mu^+ \nu_\mu$ leptonic decay of hadrons

$\mu^+ \rightarrow e^+ \nu_e + \bar{\nu}_\mu$ leptonic decay of μ

ν always present $\Rightarrow$ MET in the event.
 $\Rightarrow$ unbalanced event topology

$\Lambda(uds) \rightarrow p \pi^-$

hadronic decay

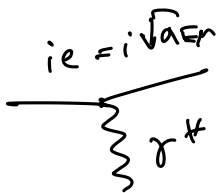
$K^+ \rightarrow \pi^+ \pi^0$

weak decays because $\Delta S = 1$

$K^0 \rightarrow \pi^+ \pi^-$

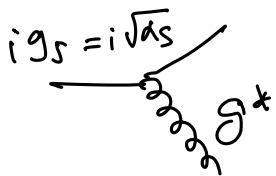
also lifetime or decay width different than hadronic interaction

Weak interaction vertex

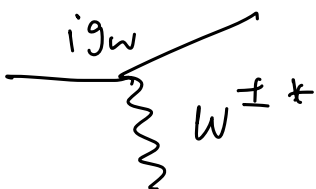


QED

$$\alpha_{EM} \approx \frac{1}{137}$$



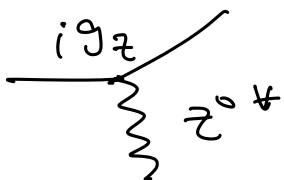
QCD $\alpha_s \approx 0.1$ (9) $Q^2 \approx 90 \text{ GeV}^2$



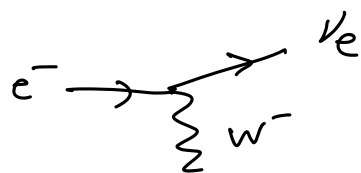
2 charged vector mesons

1 neutral vector meson

massive mediators.



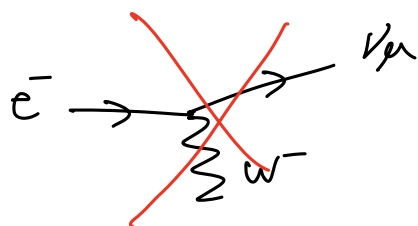
$$\alpha_W = \frac{g_W^2}{4\pi} = ?$$



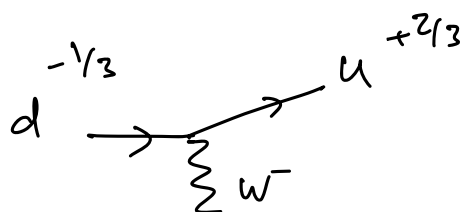
lepton # conserved.
Charge of lepton changed.



for all 3 lepton families.



lepton
Flavor changing not allowed



weak interaction: $d \rightarrow u$



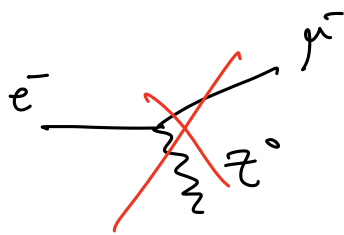
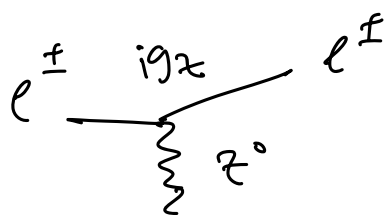
quark flavor change always with $w^\pm$
Flavor changing charged weak current

$$\Delta S = 1.$$

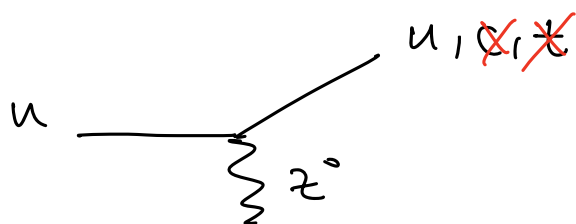
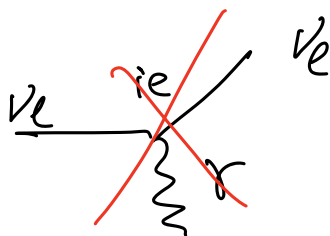
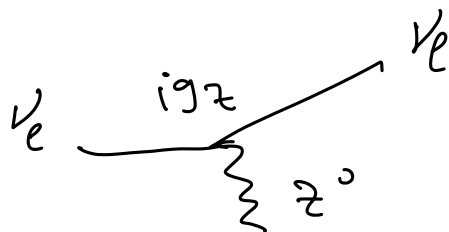
$$\Delta C = 1$$

$$\Delta b = 1$$

allowed in weak interactions

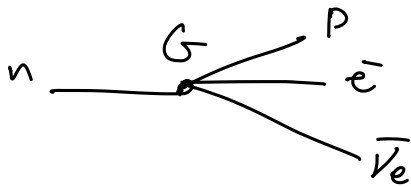


no flavor change
in neutral current



Flavor conserving
neutral current

Beta Decay



4-fermion theory

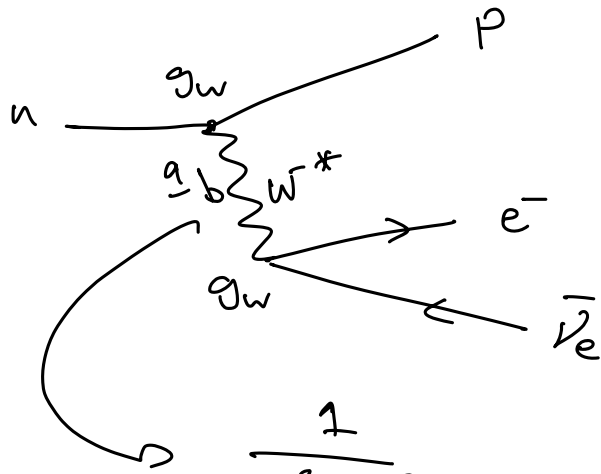
$$\mu \sim G_F$$



$$\# \text{ events} \propto G_F^2$$

Experimental meas.

$$G_F = 1.16 \times 10^{-5} \text{ GeV}^{-2}$$



$$\mu \sim \frac{g_w^2}{q^2 - m_w^2}$$

$$q^2 \sim \text{MeV}^2$$

$$m_w \approx 80 \text{ GeV} \gg q^2$$

$$\mu \sim \frac{g_w^2}{m_w^2}$$

$$\# \text{ events} \propto \left(\frac{g_w}{m_w} \right)^4$$

$$G_F \sim \frac{g_w^2}{m_w^2}$$

$$G_F = \frac{\sqrt{2}}{8} \frac{g_w^2}{m_w^2}$$

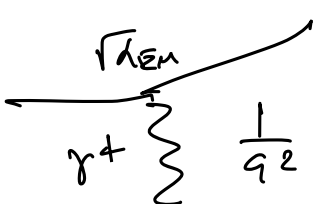
G_F, m_w measured exp.

$$\Rightarrow g_w \approx 0.65 \Rightarrow \alpha_w = \frac{g_w^2}{4\pi} \approx \frac{1}{29.5}$$

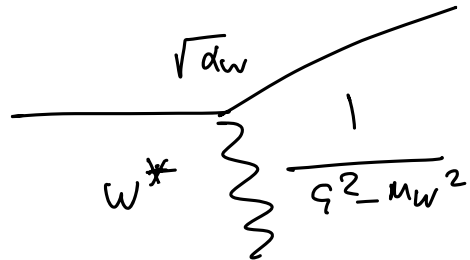
$$\alpha_{EM} \approx \frac{1}{137}$$

$$\alpha_s \approx \frac{1}{10}$$

$$\alpha_w \approx \frac{1}{29.5}$$



$$\mu \sim \frac{\alpha_{EM}}{q^2}$$



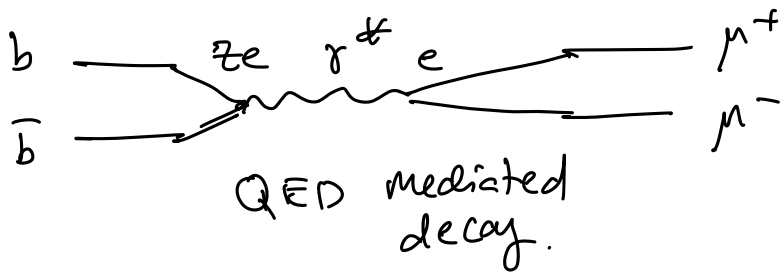
$$\mu \sim \frac{\alpha_w}{q^2 - m_w^2}$$

weak interactions are weak only $q^2 \ll m_W^2$

apparent weakness due to heavy mass of W/Z

$\Upsilon = (b\bar{b})$ bound state

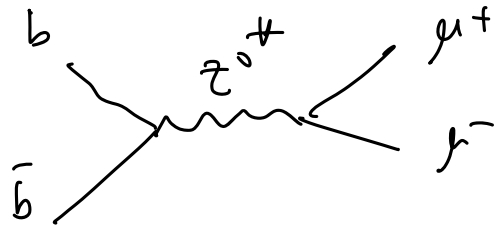
$\Upsilon \rightarrow \mu^+\mu^-$ for discovery.



$$\mu_{\text{QED}} \sim \frac{Ze^2}{q^2}$$

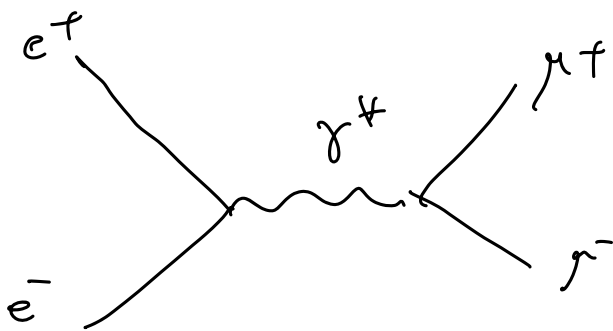
$$q^2 \simeq m_\Upsilon^2 \simeq (9 \text{ GeV})^2$$

$$\mu_{\text{QED}} \gg \mu_Z$$

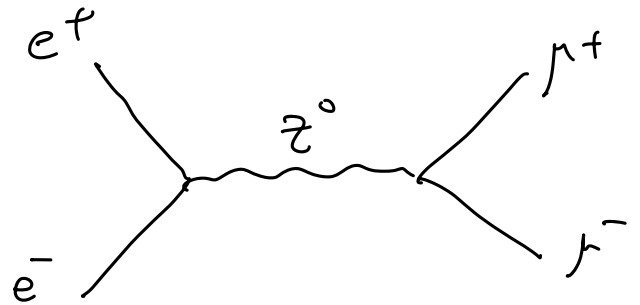


$$\mu_Z \sim \frac{g_Z^2}{q^2 - m_Z^2}$$

$$m_Z = 90 \text{ GeV}$$

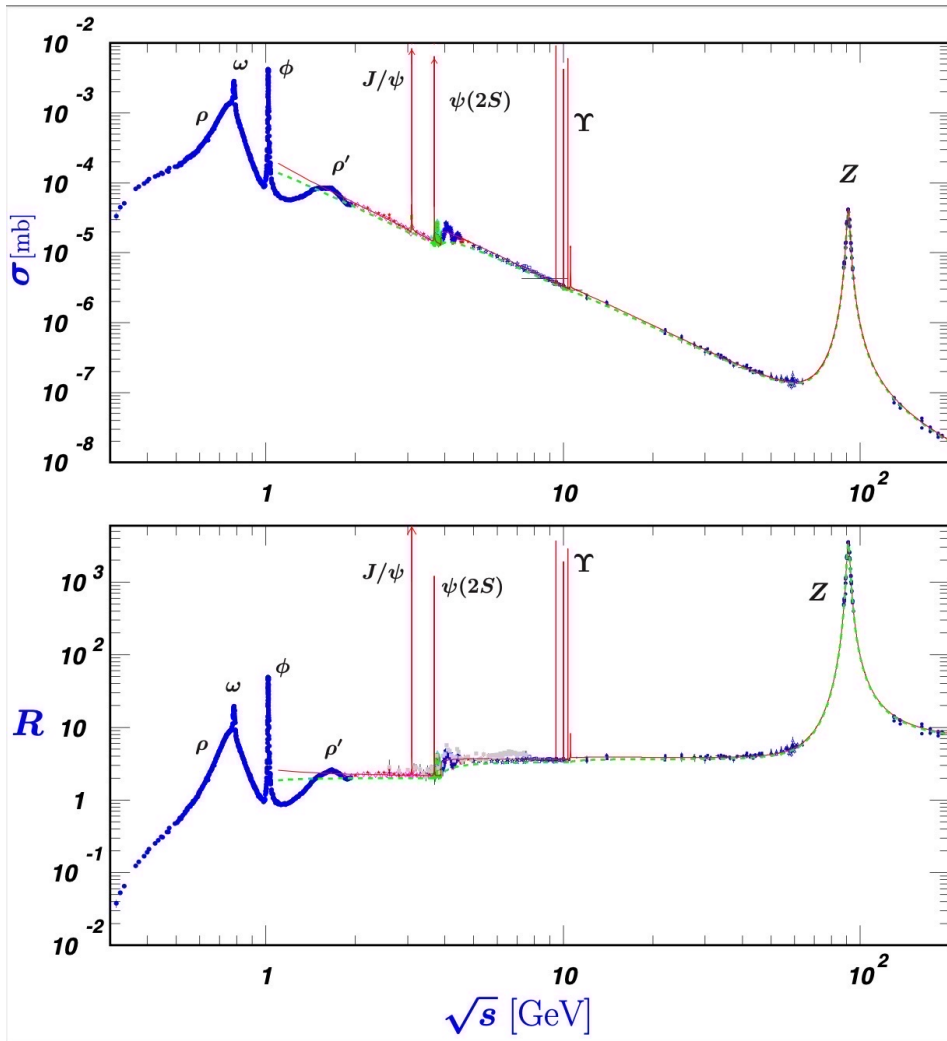


$$\mu \sim \frac{\alpha}{q^2} = \frac{\alpha}{s}$$



$$\frac{g_Z^2}{q^2 - m_Z^2} = \frac{\alpha}{s - m_Z^2}$$

$s \rightarrow m_Z^2 \Rightarrow Z$ mediation becomes dominant



z^0 becomes dominant for $s \sim m_z^2$

QED

$$ie\gamma^\mu \rightarrow \frac{-ig^{\mu\nu}}{q^2}$$

W/Z

$$\frac{ig_W}{i q^2} \rightarrow \frac{-i(g^{\mu\nu} - q^\mu q^\nu / m_{W/Z}^2)}{q^2 - m_{W/Z}^2}$$

What is the Lorentz structure of weak int. at vertex

Exp. properties

	P	C	CP
QED	✓	✓	✓
QCD	✓	✓	✓
Weak	✗	✗	✗

1957 Wu exp.

→ Goldhaber exp. ν helicity.

1964
Cronin-Fitch exp.
 $K^0 \rightarrow \pi^+ \pi^- \pi^+ \pi^-$

Goldhaber experiment:

ν only left handed.

$\bar{\nu}$ only right handed.

$\cancel{A} + \cancel{C} \Rightarrow$ affect structure of weak interactions.
 $\Rightarrow$ weak interaction depends on chirality of particles.

helicity:
$$h =: \frac{\vec{p} \cdot \vec{S}}{|\vec{p}| \cdot |\vec{S}|}$$

$\vec{p}$: mom.
 $\vec{S}$: spin.

LH: $\xrightarrow{\leftarrow} p \quad h = -1$

RH: $\xrightarrow{\rightarrow} p \quad h = +1$

massless particles: helicity is Lorentz inv.
same in all references.

massive particles: helicity can change

for ultrarelativistic particles helicity $\approx$ chirality
depends on
$$\beta = \frac{p}{E} = \frac{p}{\sqrt{p^2 + m^2}}$$